NANYANG TECHNOLOGICAL UNIVERSITY

SC2006 Software Engineering

Lab 1 Submission

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Problem Statement

Commuters in Singapore face difficulty evaluating and comparing trade-offs among multiple transport modes due to differences in travel time, reliability, crowding levels and costs, especially under dynamic real-time conditions. This makes it challenging to users to decide which transport mode best aligns with their preferences and situational constraints.

Project Mission Statement

We will develop a route recommendation application for commuters in Singapore that utilizes real-time public transport data to help users compare routes based on travel time, reliability (delay risk), and comfort (crowding). This project will be considered complete when the application has been tested and approved for release by the Land Transport Authority. The project supports Singapore's objective of improving commuting efficiency and reliability by enabling more informed travel decisions.

Functional Requirements

1. User Input Preference

1.1. First-Time User Detection

1.1.1. The system shall detect whether the User is accessing the application for the first time

1.1.2. If the User is accessing the application for the first time, the system shall prompt the user to configure their travel preferences

1.2. Initial Preference Input

1.2.1. The system shall allow the User to select a default optimisation preference during first-time access

1.2.2. The system shall provide the following optimisation preferences:

- Fastest Route
- Most Reliable Route
- Most Comfortable Route
- Most Economical Route

1.2.3. The system shall allow the user to review a brief description of each optimisation preference before selection

1.2.4 The system shall inform the User that the default preference profile can be modified later in the Settings screen

1.3. Initial Preference Constraints

1.3.1. The system shall allow the user to specify walking tolerance during initial preference setup

1.3.1.1. The system shall allow the user to select a maximum walking time (e.g., 5, 10, 15 or 20 minutes)

1.3.2. The system shall allow the user to specify transfer tolerance during initial preference setup

1.3.2.1. The system shall allow the user to select a maximum number of transfers (e.g., 0, 1, 2 or 3)

1.4. Mode Constraints

1.4.1. Allowed Transport Modes

1.4.1.1. The system shall allow the user to enable or disable transport mode categories for generation

1.4.1.2. The system shall support at minimum the following transport mode categories:

- MRT-only
- MRT and Bus
- Taxi
- Drive

1.4.1.3. The system shall allow the user to restrict route recommendations to public transport only by excluding Taxi and Drive modes

1.4.1.4. The system shall allow the User to exclude taxi routes while other transport mode categories remain enabled

1.4.1.5. The system shall require at least one transport mode category to remain enabled

1.5. Hybrid Taxi Preference

1.5.1. The system shall allow the user to enable Hybrid Taxi options

1.5.1.1. When Hybrid Taxi is enabled, the system shall allow routes that combine public transport segments with a taxi segment

1.5.1.2. The system shall allow the user to set a maximum taxi segment duration for Hybrid Taxi routes (e.g., ≤ 10 minutes)

1.5.1.3. The system shall allow the user to set a maximum taxi budget for Hybrid Taxi routes

1.6. Cost Preferences

1.6.1. Fare Eligibility

1.6.1.1. The system shall allow the user to indicate whether concession fares apply (e.g., student or senior fares)

1.6.2. Budget Constraint

1.6.2.1. If a budget is specified, the system shall avoid recommending routes that exceed the budget when feasible

1.7. Saving User Preference

1.7.1. Upon completion of the initial preference setup, the system shall save the User's selected preferences for future use

1.7.2. The system shall apply the saved preferences as the default preference profile for subsequent trip requests

2. Trip Request Input

2.1. Initiate Trip Request

2.1.1 Allow the User to initiate a new trip request

2.1.2. Retrieve the User's saved default preference profile
2.1.3. Display the active default preference profile to the User

2.2. Confirm Profile Preference Usage

2.2.1. The system asks the User whether to apply the default preference profile for the current trip

2.2.2. Allow the User to choose "Default Preferences"

2.2.3. Allow the User to choose "Modify Preferences for This Trip"

2.3. Apply Preference Configuration

2.3.1. If the User selects "Default Preferences", the system applies the saved default preference profile

2.3.2. If the User selects "Modify Preferences", the system shall apply the trip-level overrides without updating the saved default profile

2.4. Origin Input

2.4.1. Origin Input Using Current Location

2.4.1.1. The system shall allow the user to set the origin using the device's current location (GPS)

2.4.1.2. The system shall request location permission before accessing the device location

2.4.1.3. If the user denies location permission, the system shall provide an option to proceed with manual origin input

2.4.1.4. If location services are disabled, the system shall prompt the user to enable location services or use manual origin input

2.4.1.5. If the GPS location cannot be obtained within a predefined time limit, the system shall prompt the user to retry or use manual origin input

2.4.2. Manual Origin Input

2.4.2.1. The system shall allow the user to manually enter an origin location

2.4.2.2. The system shall display a list of suggested locations while the user is entering the origin

2.4.2.3. The system shall update the suggestion list when the user modifies the input text

2.4.2.4. The system shall allow the user to select a suggested location to populate the origin field

2.4.3. Origin Validation

2.4.3.1. The system shall validate that the selected or entered origin is within Singapore.

2.4.3.2. If the origin is invalid, the system shall display an error message indicating the issue

2.5. Destination Input

2.5.1. Destination Entry

2.5.1.1. The system shall allow the user to manually enter a destination location

2.5.1.2. The system shall display destination suggestions while the user is entering the destination

2.5.1.3. The system shall allow the user to select a destination from the suggestion list

2.5.1.4. The system shall allow the user to select a destination by interacting with a map

2.5.2. Destination Validation

2.5.2.1. The system shall validate that the destination is within Singapore

2.5.2.2. If the destination is invalid, the system shall display an error message and request re-entry

2.5.2.3. If the origin and destination are the same, the system shall prompt the user to re-enter the destination

2.5.3. Departure Time Options

2.5.3.1. The system shall allow the user to select the departure time as “Now”.

2.5.3.2. The system shall allow the user to schedule a departure time within the next 24 hours.

2.5.3.3. The system shall reject scheduled departure times earlier than the current time.

2.5.3.4. The system shall allow the user to edit the selected departure time before submitting the trip request.

2.5.3.5. The system shall display the selected departure time in the trip request summary prior to route generation.

3. Assess Transport Conditions and Generate Risk Scores

3.1. Initiate Transport Condition Assessment

3.1.1. After a trip request is confirmed, the system starts assessing the transport conditions

3.1.2. The system uses validated trip inputs (origin, destination, departure time) as the assessment context

3.1.3. The system applies the active preference profile, including default preferences and any trip-level overrides

3.2. Retrieve Real-Time Transport Data

3.2.1. The system retrieves real-time transport data relevant to the trip context

3.2.2. The system retrieves real-time MRT station crowdedness data

3.2.3. The system retrieves real-time MRT service and delay information

3.2.4. If Taxi or Hybrid Taxi preference option is enabled, the system retrieves real-time taxi availability

3.2.5. If Drive option is enabled, the system retrieves real-time car park availability data

3.3. Data Freshness Compliance

3.3.1 The system shall validate the freshness of real-time transport data in accordance with Section 5.1.

3.3.2 If retrieved transport data does not meet the predefined freshness criteria, the system shall treat the data as outdated in accordance with Section 5.1.3.

3.4. Fallback Data Compliance

3.4.1 If real-time transport data is unavailable or outdated, the system shall apply fallback handling in accordance with Section 5.2.

3.4.2 The system shall mark any data elements operating in fallback mode in accordance with Section 5.3.

3.5. Assess Crowding Risk

3.5.1 The system shall compute a segment-level crowding risk indicator for relevant public transport segments based on retrieved transport data.

3.5.2 The system shall categorise each segment-level crowding risk indicator as Low, Medium, or High.

3.5.3 If crowding data is unavailable for a segment, the system shall mark the segment-level crowding risk indicator as Unknown.

3.6. Assess Delay Risk

3.6.1 The system shall compute a segment-level delay risk indicator for relevant public transport segments based on retrieved transport data.

3.6.2 The system shall categorise each segment-level delay risk indicator as Low, Medium, or High.

3.6.3 If delay data is unavailable for a segment, the system shall mark the segment-level delay risk indicator as Unknown.

3.7 Prepare Risk Assessment Output

3.7.1 The system shall package the computed segment-level crowding risk indicators and segment-level delay risk indicators as the risk assessment output.

3.7.2 The system shall include, for each segment-level risk indicator, its risk category (Low/Medium/High/Unknown).

3.7.3 The system shall indicate whether each segment-level risk indicator is derived from real-time data or fallback data.

3.7.4 The system shall provide the risk assessment output as input to route generation, route-level risk aggregation, and ranking.

4. Route Generation and Ranking

4.1. Initial Route Generation

4.1.1. After crowding risk and delay risk scores have been assessed, the system shall initiate route generation

4.1.2. The system shall use the validated trip inputs (origin, destination and departure time) for route generation

4.1.3 The system shall use the segment-level crowding and delay risk indicators as inputs to route generation, route-level risk aggregation, and ranking.

4.1.4. The system shall apply the active preference profile, including default preferences and any trip-level overrides during route generation

4.2. Route Generation by Transport Mode

4.2.1. The system shall generate route options only using transport modes enabled in the active preference profile

4.2.2. The system shall generate MRT-only route options when the MRT-only mode is enabled

4.2.3. The system shall generate combined MRT and bus route when the MRT and Bus mode is enabled

4.2.4. The system shall generate Taxi-only route options when Taxi mode is enabled

4.2.5. The system shall generate Drive-only route options when Drive mode is enabled

4.2.6. If the user has restricted route recommendations to public transport only, the system shall exclude Taxi and Drive route options

4.3. Hybrid Taxi Route Generation

4.3.1. If Hybrid Taxi is enabled, the system shall generate route options that combine public transport segments with a taxi segment

4.3.2. The system shall ensure that Hybrid Taxi routes comply with the specified hybrid constraints

4.3.2.1. The system shall avoid generating Hybrid Taxi routes that exceed maximum taxi segment duration when specified

4.3.2.2. The system shall avoid generating Hybrid Taxi routes that exceed the specified budget when specified

4.4. Route Feasibility and Constraint Handling

4.4.1. The system shall generate only feasible routes that allow travel from the specified origin to the destination

4.4.2. The system shall consider the User's walking tolerance during route generation

4.4.2.1. If a maximum tolerance is specified, the system shall avoid generating routes exceeding the walking limit when feasible

4.4.3. The system shall consider the User's transfer tolerance during route generation

4.4.3.1. If a maximum number of transfers is specified, the system shall avoid generating routes exceeding the transfer limit when feasible

4.4.4. If a maximum budget is specified, the system shall avoid generating routes that exceed the budget when feasible

4.5. Route Attribute Assignment

4.5.1. For each generated route, the system shall assign an estimated travel time

4.5.2 For each generated route, the system shall compute a route-level crowding risk level (Low/Medium/High/Unknown) by aggregating segment-level crowding risk indicators across the segments of the route

4.5.3 For each generated route, the system shall compute a route-level delay risk level (Low/Medium/High/Unknown) by aggregating segment-level delay risk indicators across the segments of the route

4.5.4. For each generated route, the system shall determine the number of transfers involved

4.5.5. For each generated route, the system shall determine the estimated travel cost when applicable

4.6. Route Ranking According to Profile Preference

4.6.1. The system shall rank generated routes according to the optimisation goal in the active profile preference

4.6.2. Ranking by Fastest Route

4.6.2.1 If the optimisation goal is Fastest Route, the system shall prioritise routes with shorter estimated travel times

4.6.3. Ranking by Most Reliable Route

4.6.3.1. If the optimisation goal is Most Reliable Route, the system shall prioritise routes with lower route-level delay risk scores

4.6.3.2. If multiple routes have similar delay risk scores, the system shall further prioritise routes with lower route-level crowding risk scores

4.6.4. Ranking by Most Comfortable Route

4.6.4.1. If the optimisation goal is Most Comfortable Route, the system shall prioritise routes with lower route-level crowding risk scores

4.6.4.2. If multiple routes have similar crowding risk scores, the system shall further prioritise routes with fewer transfers and shorter waiting times

4.6.5. Ranking by Most Economical Route

4.6.5.1 If the optimisation goal is Most Economical Route, the system shall prioritise routes with lower estimated travel cost.

4.6.5.2 If multiple routes have similar estimated cost, the system shall prioritise routes with shorter estimated travel time.

4.6.5.3 If multiple routes remain tied, the system shall prioritise routes with fewer transfers.

4.7. Route Selection Output

4.7.1. The system shall generate between two and three ranked route options for each trip request.

4.7.2. Each generated route option shall represent a distinct travel alternative.

4.7.3. If fewer than two feasible routes can be generated, the system shall notify the user accordingly.

4.8. Route Explanation

4.8.1. For each generated route option, the system shall provide a brief one-line explanation.

4.8.2. The explanation shall describe how the route aligns with the User's selected optimisation goal and the route's route-level risk levels.

5. Data Freshness, Fallback & Settings

5.1. Data Freshness Management

5.1.1. The system shall associate a timestamp with each retrieved real-time transport data source

5.1.2. The system shall determine whether retrieved transport data is sufficiently recent based on predefined freshness criteria

5.1.3. If retrieved transport data does not meet freshness criteria, the system shall treat the data as outdated

5.2. Fallback Data Handling

5.2.1. If real-time transport data is unavailable or outdated, the system shall use the most recent available transport data for processing

5.2.2. The system shall identify which transport sources are operating in fallback mode

5.2.3. The system shall ensure that fallback data is used consistently during risk assessment and route generation

5.2.4. The system shall prevent partial mixing of real-time and outdated data for the same transport mode within a single trip request

5.3. Impact Awareness on Route Recommendations

5.3.1. If fallback data is used during risk assessment or route generation, the system shall mark the affected route recommendations accordingly

5.3.2. The system shall indicate that crowding risk or delay risk scores may be less accurate when fallback data is used

5.3.3. The system shall allow route generation and ranking to proceed when fallback data is used, unless no relevant data is available

5.4. User-Initiated Data Refresh

5.4.1. The system shall allow the User to manually refresh route recommendations before the trip starts

5.4.2. Upon refresh, the system shall re-initiate transport condition assessment using the most recent available data

5.4.3. The system shall regenerate and re-rank route options after a successful data refresh

5.5 User Settings & Localisation

5.5.1 The system shall allow the User to access the Settings screen.

5.5.2 The system shall allow the User to select a display language.

5.5.3 The system shall apply the selected language to all user-facing text, including labels, error messages, and help messages.

5.5.4 The system shall save the selected language for future sessions.

5.5.5 The system shall allow the User to reset settings to default values.

5.5.6 The system shall allow the User to select a preferred unit format (e.g., minutes, km).

5.5.7 The system shall allow the User to view the saved default preference profile in the Settings screen.

5.5.8 The system shall allow the User to update the saved default preference profile in the Settings screen.

5.5.9 Upon saving changes, the system shall apply the updated default preference profile to subsequent trip requests.

Non-Functional Requirements

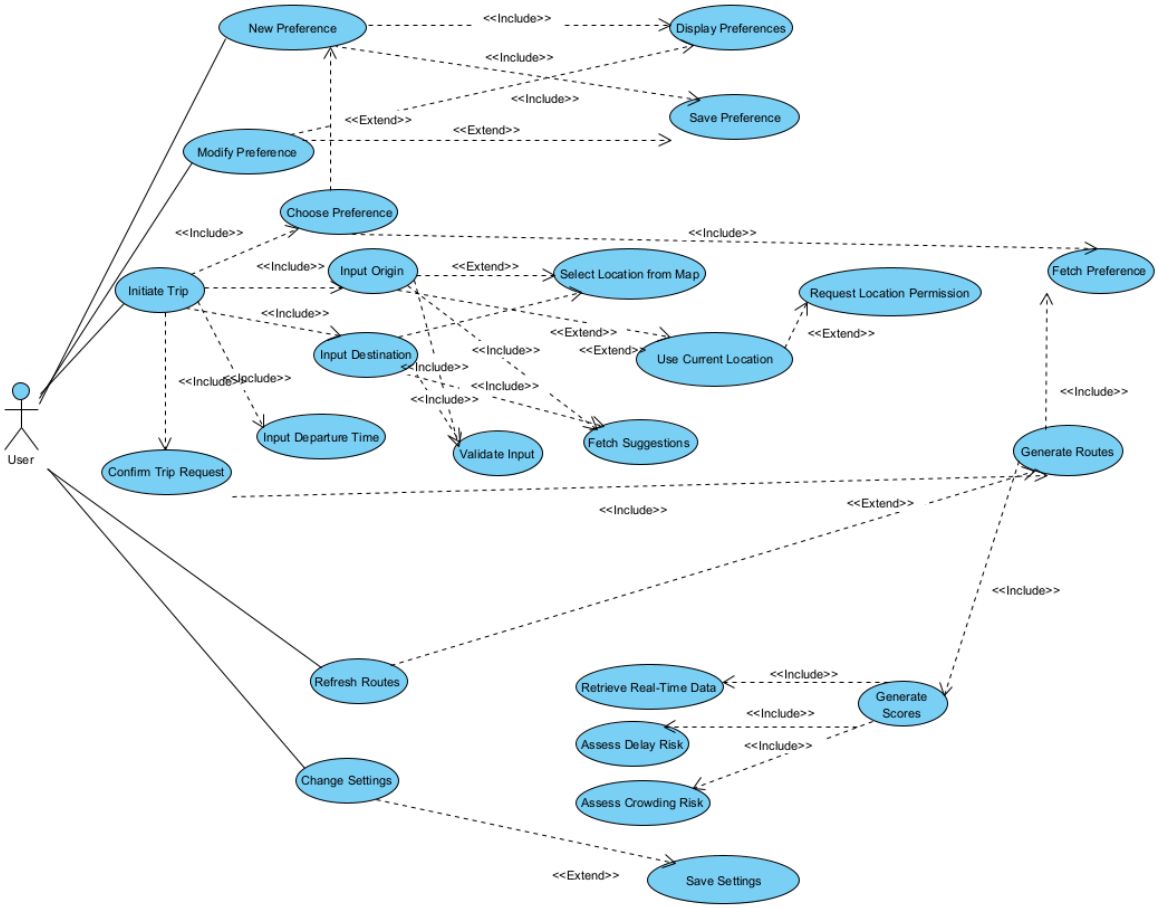
Performance	The system shall return route recommendations within 5 seconds after the user submits a valid origin and destination under normal network conditions. The system shall refresh real-time transport data at intervals of no more than 5 minutes.
Reliability	The system shall continue to provide route recommendations using the most recently cached data when one or more real-time data sources are unavailable.
Usability	The system should display a clear visual comparison of routes in a manner that allows users to visually distinguish differences in travel time, reliability and comfort. The system shall display help messages in the user's selected language.
Supportability	The system shall support at least 200 concurrent users during peak commuting hours. Under the above load, the system shall return route recommendations within 5 seconds after the user submits a valid origin and destination.
Scalability	The system shall allow new transport modes to be added by extending the route-generation module and updating configuration, without changing existing functional requirements or user workflows.
Availability & Resilience	The system shall remain usable and allow route search even when one or more data sources are unavailable, by falling back to cached data. The system shall display the timestamp of the latest successfully retrieved data on the results screen. If cached data is older than 15 minutes, the system shall show a warning that results may be less accurate.

Data Dictionary

Term	Definition
User	A commuter who uses the application to plan and evaluate travel options
User Preference Profile	A set of travel preferences defined by the User that describes how they prefer routes to be recommended, based on priorities such as travel time, reliability, comfort and cost
Optimisation Preference	The primary travel goal selected by the User to guide route recommendations
Walking Tolerance	The maximum amount of walking time the User is willing to accept as part of their trip
Transfer Tolerance	The maximum number of transfers between transport services that a User is willing to accept during a trip
Transport Mode	A travel method or mode category used to complete a trip, such as MRT, bus, taxi, drive, or combinations (e.g., MRT+Bus)
Mode Constraint	A restriction defined by the User that enables or disables specific transport modes during route generation
Hybrid Taxi Route	A travel route that combines public transport segments with a short taxi segment to improve convenience or reliability
Fare Eligibility	A condition indicating whether the User qualifies for concession fares such as student or senior discounts
Trip Budget	The maximum total cost that the User is willing to pay for a trip
Trip Request	A request made by the User to plan a journey, including origin, destination, departure time, and applied preferences
Origin	The starting location of a trip which the User begins their journey
Destination	The end location of a trip which the User intends to travel
Departure Time	The time at which the User plans to begin their journey
Real-Time Transport Data	Information reflecting current transport conditions such as crowding levels, delays, taxi availability and car park availability
Transport Segment	A continuous portion of a trip that uses a single transport service
Segment-Level Risk Indicator	An assessment of crowding or delay conditions associated with an individual transport segment within a route
Crowding Level	An indication of how crowded a transport service, such as MRT platforms or buses, is at a given moment
Delay Status	An indication of whether transport services are operating normally or experiencing delays
Crowding Risk	An assessment of the likelihood that the User will experience crowded conditions during a trip. It will be categorised as Low, Medium or High
Delay Risk	An assessment of the likelihood that the User will experience delays during a trip. It will be categorised as Low, Medium or High
Unknown Risk	A risk status indicating that sufficient data is not available to determine crowding or delay conditions for a transport segment or route

Route-Level Risk	An overall assessment of crowding or delay conditions for an entire route, derived from the risks of its constituent transport segments
Route Option	A possible travel path generated by the system that allows the User to travel from the origin to the destination using one or more transport modes according to their preferences
Route Ranking	The ordering of route options based on how well they align with the User's selected optimisation preference
Route Explanation	A brief one-line description explaining why a particular route option is recommended based on User preference and assessed transport risks
Data Freshness	A measure of how recent transport data is at the time route recommendations are generated
Fallback Data	Previously retrieved transport data that is used when real-time data is unavailable or outdated
User Settings	A collection of user-configurable options such as language, unit format and default preference profile
Localisation	The adaption of system content, including language and units of measurement, to match the user's selected preferences

Use Case Diagram



Use Case Description

Use Case ID:	1		
Use Case Name:	Configure New Preference Profile		
Created By:	Huynh Thao Tuong Van	Last Updated By:	Huynh Thao Tuong Van
Date Created:	11/2/2026	Date Last Updated:	11/2/2026

Actor:	User
Description:	The User configures a default travel preference profile, including optimisation goal and travel constraints, which will be applied automatically to subsequent trip requests.
Preconditions:	User has launched the application. No default preference profile exists OR user has chosen to update the default profile via Settings.
Postconditions:	System saves the user preferences
Priority:	High
Frequency of Use:	Low
Flow of Events:	1. System displays available preferences for the user to select using the “Display Preference” use case. 2. User selects preferences and constraints 3. System saves the users preferences
Alternative Flows:	NIL
Exceptions:	If User does not select an option, system displays an error and requires at least one option to be selected before saving.
Includes:	Display Preferences Save Preferences
Special Requirements:	Preference profile must be saved persistently across sessions.
Assumptions:	User is planning trips within Singapore. Validation rules for constraints are predefined.
Notes and Issues:	NIL

Use Case ID:	2		
Use Case Name:	Modify Preference Profile		
Created By:	Huynh Thao Tuong Van	Last Updated By:	Huynh Thao Tuong Van
Date Created:	11/2/2026	Date Last Updated:	11/2/2026

Actor:	User
Description:	Allows the user to select and update an existing preference profile
Preconditions:	At least one preference profile exists in the system
Postconditions:	System saves the user preferences
Priority:	Medium
Frequency of Use:	Low
Flow of Events:	1. System displays available preferences for the user to select using the "Display Preference" use case. 2. User selects preferences and constraints 3. System saves the users preferences
Alternative Flows:	**Cancel Modification** User exits the edit screen and system reverts to the previous menu without saving changes
Exceptions:	**Save Failure** System fails to write to the database; user is notified and prompted to retry.
Includes:	Display Preferences Save Preferences
Special Requirements:	NIL
Assumptions:	User is planning trips within Singapore. Validation rules for constraints are predefined.
Notes and Issues:	NIL

Use Case ID:	3		
Use Case Name:	Display Preferences Profile		
Created By:	Huynh Thao Tuong Van	Last Updated By:	Huynh Thao Tuong Van
Date Created:	11/2/2026	Date Last Updated:	11/2/2026

Actor:	User
Description:	Retrieves and displays current preference settings to the user
Preconditions:	Preference data exists or default settings are available
Postconditions:	Preferences are visible on the user interface
Priority:	High
Frequency of Use:	High
Flow of Events:	1. System retrieves preference data from storage. 2. System formats and renders the settings for the user interface.
Alternative Flows:	NIL
Exceptions:	**Data Unavailable** System displays default settings or a "No preferences found" message.
Includes:	NIL
Special Requirements:	NIL
Assumptions:	NIL
Notes and Issues:	NIL

Use Case ID:	4		
Use Case Name:	Save Preferences Profile		
Created By:	Huynh Thao Tuong Van	Last Updated By:	Huynh Thao Tuong Van
Date Created:	11/2/2026	Date Last Updated:	11/2/2026

Actor:	User
Description:	Saves the user's preference settings to the system database
Preconditions:	New preference data or modifications have been provided by the user
Postconditions:	Preferences are successfully stored and available for future retrieval
Priority:	High
Frequency of Use:	High
Flow of Events:	1. System receives the finalized preference data. 2. System writes the data to the database. 3. System confirms the successful save to the parent use case.
Alternative Flows:	NIL
Exceptions:	**Storage Error** System fails to write data and returns an error message to the parent use case.
Includes:	NIL
Special Requirements:	NIL
Assumptions:	Storage is not full and connection to storage exists
Notes and Issues:	NIL

Use Case ID:	5		
Use Case Name:	Initiate Trip		
Created By:	Huynh Thao Tuong Van	Last Updated By:	Huynh Thao Tuong Van
Date Created:	11/2/2026	Date Last Updated:	11/2/2026

Actor:	User
Description:	Begins the process of planning a journey by providing necessary travel parameters
Preconditions:	NIL
Postconditions:	Travel parameters are captured and ready for route generation
Priority:	High
Frequency of Use:	High
Flow of Events:	1. System prompts for origin, destination, and departure time. 2. User inputs the origin (**Input Origin**). 3. User inputs the destination (**Input Destination**). 4. User inputs the departure time (**Input Departure Time**). 5. System prompts the user to select specific preferences (**Choose Preference**). 6. User confirms the trip details (**Confirm Trip Request**).
Alternative Flows:	NIL
Exceptions:	**Invalid Input** System detects missing or incompatible locations and prompts for correction (Include: Validate Input).
Includes:	Input Origin Input Destination Input Departure Time Choose Preference Confirm Trip Request Validate Input
Special Requirements:	NIL
Assumptions:	The system has access to a map API and a database of transport schedules
Notes and Issues:	NIL

Use Case ID:	6		
Use Case Name:	Choose Preference		
Created By:	Huynh Thao Tuong Van	Last Updated By:	Huynh Thao Tuong Van
Date Created:	11/2/2026	Date Last Updated:	11/2/2026

Actor:	User
Description:	Enables the user to select a specific preference profile to be applied to their trip
Preconditions:	User has initiated a trip
Postconditions:	A preference profile is active for the current session
Priority:	High
Frequency of Use:	High
Flow of Events:	1. System retrieves available preference profiles (**Fetch Preference**). 2. System presents profiles to the user (**Display Preferences**). 3. User selects the preferred profile (Default or Modify Preferences)
Alternative Flows:	**Create New** User opts to define a new set of preferences (**New Preference**) **Modify Existing:** User opts to edit a profile before selection (**Modify Preference**).
Exceptions:	NIL
Includes:	Fetch Preference Display Preference
Special Requirements:	NIL
Assumptions:	A “default” preference profile exists
Notes and Issues:	NIL

Use Case ID:	7		
Use Case Name:	Input Origin		
Created By:	Huynh Thao Tuong Van	Last Updated By:	Huynh Thao Tuong Van
Date Created:	11/2/2026	Date Last Updated:	11/2/2026

Actor:	User
Description:	Captures the starting point of the journey through manual entry or automated location services
Preconditions:	User has initiated a trip
Postconditions:	Origin coordinates or address are successfully defined
Priority:	High
Frequency of Use:	High
Flow of Events:	1. System prompts the user for a starting location. 2. User enters text address or location name. 3. System validates the input (**Validate Input**). 4. System provides location suggestions based on the input (**Fetch Suggestions**)
Alternative Flows:	**Map Selection** User picks a point directly on the interface (Extend: Select Location from Map). **GPS Positioning** User uses their current device location (Extend: Use Current Location).
Exceptions:	**Unresolved Address** System cannot find the location and prompts user to refine search or use the map.
Includes:	Validate Input Fetch Suggestions
Special Requirements:	NIL
Assumptions:	User has a stable internet connection for location fetching
Notes and Issues:	NIL

Use Case ID:	8		
Use Case Name:	Input Destination		
Created By:	Huynh Thao Tuong Van	Last Updated By:	Huynh Thao Tuong Van
Date Created:	11/2/2026	Date Last Updated:	11/2/2026

Actor:	User
Description:	Captures the ending point of the journey through manual entry or automated location services
Preconditions:	User has initiated a trip
Postconditions:	Ending coordinates or address are successfully defined
Priority:	High
Frequency of Use:	High
Flow of Events:	1. System prompts the user for a ending location. 2. User enters text address or location name. 3. System validates the input (**Validate Input**). 4. System provides location suggestions based on the input (**Fetch Suggestions**)
Alternative Flows:	**Map Selection** User picks a point directly on the interface (Extend: Select Location from Map).
Exceptions:	**Unresolved Address** System cannot find the location and prompts user to refine search or use the map.
Includes:	Validate Input Fetch Suggestions
Special Requirements:	NIL
Assumptions:	User has a stable internet connection for location fetching
Notes and Issues:	NIL

Use Case ID:	9		
Use Case Name:	Select Location from Map		
Created By:	Huynh Thao Tuong Van	Last Updated By:	Huynh Thao Tuong Van
Date Created:	11/2/2026	Date Last Updated:	11/2/2026

Actor:	User
Description:	Allows the user to define a location by interacting with a visual map interface
Preconditions:	The map interface is loaded and accessible
Postconditions:	A specific geographic point is captured and assigned to the origin or destination
Priority:	Medium
Frequency of Use:	High
Flow of Events:	1. User triggers the map selection option. 2. System opens an interactive map view. 3. User pans, zooms, and clicks/taps to select a specific point. 4. System identifies the location/address of the selected point. 5. User confirms the selection.
Alternative Flows:	**Cancel Selection** User exits the map view; system returns to the manual input field without updating data.
Exceptions:	**Map Rendering Error** System fails to load tiles; user is prompted to use manual text input. **Out of Bounds** Selected point is in an unsupported region; system requests a different selection.
Includes:	NIL
Special Requirements:	NIL
Assumptions:	User has a stable internet connection for location fetching
Notes and Issues:	NIL

Use Case ID:	10		
Use Case Name:	Use Current Location		
Created By:	Huynh Thao Tuong Van	Last Updated By:	Huynh Thao Tuong Van
Date Created:	11/2/2026	Date Last Updated:	11/2/2026

Actor:	User
Description:	Automatically detects and retrieves the user's real-time geographic coordinates to populate trip parameters
Preconditions:	Device has location services (GPS/GNSS) and network connectivity enabled
Postconditions:	Current location is captured and assigned as the origin
Priority:	High
Frequency of Use:	High
Flow of Events:	1. User selects the "Use Current Location" option within an input field. 2. System accesses the device's location sensor. 3. System retrieves the current latitude and longitude. 4. System resolves the coordinates to a valid location or address.
Alternative Flows:	**Permission Request** If access is not already granted, the system triggers the permission flow (**Request Location Permission**)
Exceptions:	**Signal Loss** System cannot acquire a location fix; user is prompted to enter the location manually. **Location Services Disabled** Device GPS is off; system requests the user to enable it in settings.
Includes:	NIL
Special Requirements:	NIL
Assumptions:	NIL
Notes and Issues:	NIL

Use Case ID:	11		
Use Case Name:	Request Location Permission		
Created By:	Huynh Thao Tuong Van	Last Updated By:	Huynh Thao Tuong Van
Date Created:	11/2/2026	Date Last Updated:	11/2/2026

Actor:	User
Description:	Prompts the user to grant the application access to the device's geographic location data
Preconditions:	The application does not have active permission to access location services
Postconditions:	Permission is either granted, allowing location retrieval, or denied, blocking the feature
Priority:	Medium
Frequency of Use:	Low
Flow of Events:	1. System triggers the OS-level permission dialog. 2. User reviews the request. 3. User selects "Allow" or "Allow while using app." 4. System registers the permission and proceeds with location retrieval.
Alternative Flows:	**Permission Denied** User selects "Do not allow"; system notifies the user that the feature is unavailable without permission
Exceptions:	**OS-Level Restriction** Location services are disabled entirely in device settings; system prompts user to enable them in the OS.
Includes:	NIL
Special Requirements:	NIL
Assumptions:	NIL
Notes and Issues:	NIL

Use Case ID:	12		
Use Case Name:	Input Departure Time		
Created By:	Ong Zhi Jie	Last Updated By:	Ong Zhi Jie
Date Created:	11/2/2026	Date Last Updated:	11/2/2026

Actor:	User
Description:	Captures the desired start time for the trip to allow for scheduled route calculations
Preconditions:	User has initiated a trip
Postconditions:	A valid timestamp is stored for the trip request
Priority:	High
Frequency of Use:	High
Flow of Events:	1. System prompts the user for a departure time. 2. User selects or inputs the date and time.
Alternative Flows:	**Leave Now** User selects an immediate departure option; system applies the current system time.
Exceptions:	**Past Time** User enters a time that has already passed; system prompts for a future time.
Includes:	NIL
Special Requirements:	NIL
Assumptions:	NIL
Notes and Issues:	NIL

Use Case ID:	13		
Use Case Name:	Confirm Trip Request		
Created By:	Ong Zhi Jie	Last Updated By:	Ong Zhi Jie
Date Created:	11/2/2026	Date Last Updated:	11/2/2026

Actor:	User
Description:	Finalise the trip input process and trigger system to calculate and display travel options
Preconditions:	Origin, destination, and departure time have been successfully provided and validated
Postconditions:	The system initiates the route calculation process
Priority:	High
Frequency of Use:	High
Flow of Events:	1. System presents a summary of all trip parameters for user review. 2. User selects the confirmation button. 3. System triggers the route calculation logic (**Generate Routes**).
Alternative Flows:	NIL
Exceptions:	NIL
Includes:	Generate Routes
Special Requirements:	NIL
Assumptions:	NIL
Notes and Issues:	NIL

Use Case ID:	14		
Use Case Name:	Fetch Preferences		
Created By:	Ong Zhi Jie	Last Updated By:	Ong Zhi Jie
Date Created:	11/2/2026	Date Last Updated:	11/2/2026

Actor:	User
Description:	Retrieves stored preference profiles from the system database for user selection
Preconditions:	System is connected to the database
Postconditions:	All available preference data is retrieved and passed to the display component
Priority:	High
Frequency of Use:	High
Flow of Events:	1. System queries the database for profiles associated with the user account. 2. System retrieves transport, cost, and speed preferences. 3. System sends the data to the **Choose Preferences** use case.
Alternative Flows:	NIL
Exceptions:	NIL
Includes:	NIL
Special Requirements:	NIL
Assumptions:	NIL
Notes and Issues:	NIL

Use Case ID:	15		
Use Case Name:	Validate Input		
Created By:	Ong Zhi Jie	Last Updated By:	Ong Zhi Jie
Date Created:	11/2/2026	Date Last Updated:	11/2/2026

Actor:	User
Description:	Verifies that the data entered by the user (origin or destination) is logically consistent and in the correct format
Preconditions:	Data has been entered into an input field
Postconditions:	Input is either accepted as valid or flagged for correction
Priority:	High
Frequency of Use:	High
Flow of Events:	1. System receives raw input data from a parent use case. 2. System verifies that locations are valid. 3. System returns a "Valid" status to the parent flow.
Alternative Flows:	NIL
Exceptions:	**Validation Failure** System identifies errors; returns "Invalid" status and error messages to the user.
Includes:	NIL
Special Requirements:	NIL
Assumptions:	NIL
Notes and Issues:	NIL

Use Case ID:	16		
Use Case Name:	Fetch Suggestions		
Created By:	Ong Zhi Jie	Last Updated By:	Ong Zhi Jie
Date Created:	11/2/2026	Date Last Updated:	11/2/2026

Actor:	User
Description:	Provides real-time address or landmark suggestions based on partial text input
Preconditions:	Data has been entered into an input field
Postconditions:	A list of relevant matching locations is displayed
Priority:	Medium
Frequency of Use:	High
Flow of Events:	1. System detects text input in the location field. 2. System queries the map database for matching strings. 3. System retrieves and ranks relevant suggestions. 4. System displays the list to the user for selection.
Alternative Flows:	NIL
Exceptions:	**No Matches** System displays "No results found" or shows no list. **API Timeout** System fails to fetch data; user must continue typing manually.
Includes:	NIL
Special Requirements:	NIL
Assumptions:	NIL
Notes and Issues:	NIL

Use Case ID:	17		
Use Case Name:	Generate Route		
Created By:	Ong Zhi Jie	Last Updated By:	Ong Zhi Jie
Date Created:	11/2/2026	Date Last Updated:	11/2/2026

Actor:	User
Description:	Computes and ranks possible travel paths based on confirmed trip data and preferences
Preconditions:	Origin and destination and departure time are confirmed
Postconditions:	A list of scored routes is presented to the user
Priority:	High
Frequency of Use:	High
Flow of Events:	1. System retrieves active preference settings (**Fetch Preference**). 2. System identifies multiple potential route candidates. 3. System evaluates and assigns scores to each candidate (**Generate Scores**). 4. System displays the ranked list of routes to the user.
Alternative Flows:	NIL
Exceptions:	**No Route Found** System notifies the user if no viable paths exist between the chosen locations.
Includes:	Fetch Preferences Generate Scores
Special Requirements:	NIL
Assumptions:	NIL
Notes and Issues:	NIL

Use Case ID:	18		
Use Case Name:	Generate Scores		
Created By:	Ong Zhi Jie	Last Updated By:	Ong Zhi Jie
Date Created:	11/2/2026	Date Last Updated:	11/2/2026

Actor:	System
Description:	Calculates a numerical ranking for each potential route by evaluating real-time variables and risk factors
Preconditions:	A set of candidate routes has been identified
Postconditions:	Each route is assigned a score for ranking and display
Priority:	High
Frequency of Use:	High
Flow of Events:	1. System pulls current traffic and transport status (**Retrieve Real-Time Data**). 2. System evaluates potential time losses (**Assess Delay Risk**). 3. System evaluates passenger density levels (**Assess Crowding Risk**). 4. System applies weighted formulas based on user preferences to calculate the final score
Alternative Flows:	NIL
Exceptions:	**Data Timeout** If real-time data is unavailable, the system generates scores based on historical averages and notifies the user of the data age.
Includes:	Retrieve Real-Time Data Assess Delay Risk Assess Crowding Risk
Special Requirements:	NIL
Assumptions:	NIL
Notes and Issues:	NIL

Use Case ID:	19		
Use Case Name:	Retrieve Real-Time Data		
Created By:	Ong Zhi Jie	Last Updated By:	Ong Zhi Jie
Date Created:	11/2/2026	Date Last Updated:	11/2/2026

Actor:	System
Description:	Fetches current status information for transport modes, traffic conditions, and service alerts from external providers
Preconditions:	System has an active network connection to external data APIs
Postconditions:	Up-to-the-minute transport data is available for scoring and risk assessment
Priority:	High
Frequency of Use:	High
Flow of Events:	1. System identifies the geographic area and transport modes involved in the candidate routes. 2. System sends requests to external real-time data providers. 3. System receives and parses incoming data (e.g., traffic speeds, vehicle locations, service delays). 4. System passes the processed data to the scoring engine.
Alternative Flows:	NIL
Exceptions:	**API Connection Failure** System cannot reach the provider; it falls back to historical data. **Malformed Data** Received data is unreadable; system discards it and attempts a retry or uses default values.
Includes:	None
Special Requirements:	NIL
Assumptions:	NIL
Notes and Issues:	NIL

Use Case ID:	20		
Use Case Name:	Assess Delay Risk		
Created By:	Ong Zhi Jie	Last Updated By:	Ong Zhi Jie
Date Created:	11/2/2026	Date Last Updated:	11/2/2026

Actor:	System
Description:	Evaluates the probability and potential impact of travel delays for a specific route based on live and historical data
Preconditions:	Real-time transport data and historical performance data are available
Postconditions:	A delay risk factor is calculated and provided to the scoring engine
Priority:	High
Frequency of Use:	High
Flow of Events:	1. System analyzes current service alerts and traffic congestion levels. 2. System compares current travel times against historical averages for the specific time of day. 3. System identifies bottlenecks or high-incident zones along the route. 4. System calculates a risk percentage or time-buffer requirement.
Alternative Flows:	NIL
Exceptions:	NIL
Includes:	None
Special Requirements:	NIL
Assumptions:	NIL
Notes and Issues:	NIL

Use Case ID:	21		
Use Case Name:	Assess Crowding Risk		
Created By:	Ong Zhi Jie	Last Updated By:	Ong Zhi Jie
Date Created:	11/2/2026	Date Last Updated:	11/2/2026

Actor:	System
Description:	Evaluates the probability and potential impact of public transport crowding for a specific route based on live and historical data
Preconditions:	Real-time transport data and historical performance data are available
Postconditions:	A delay risk factor is calculated and provided to the scoring engine
Priority:	High
Frequency of Use:	High
Flow of Events:	1. System analyzes current service alerts and human congestion levels. 2. System compares current travel times against historical averages for the specific time of day. 3. System identifies bottlenecks or high-incident zones along the route. 4. System calculates a risk percentage or time-buffer requirement.
Alternative Flows:	NIL
Exceptions:	NIL
Includes:	None
Special Requirements:	NIL
Assumptions:	NIL
Notes and Issues:	NIL

Use Case ID:	22		
Use Case Name:	Refresh Routes		
Created By:	Ong Zhi Jie	Last Updated By:	Ong Zhi Jie
Date Created:	11/2/2026	Date Last Updated:	11/2/2026

Actor:	User
Description:	Manually triggers an update of the calculated routes to account for the most recent real-time data and traffic conditions
Preconditions:	A set of routes has already been generated and is currently displayed to the user
Postconditions:	The route list is updated with current scores, times, and risk assessments
Priority:	Medium
Frequency of Use:	High
Flow of Events:	1. User selects the "Refresh" option on the results screen. 2. System re-triggers the route ranking logic (**Generate Routes**). 3. System replaces the existing results with the updated list.
Alternative Flows:	NIL
Exceptions:	NIL
Includes:	None
Special Requirements:	NIL
Assumptions:	NIL
Notes and Issues:	NIL

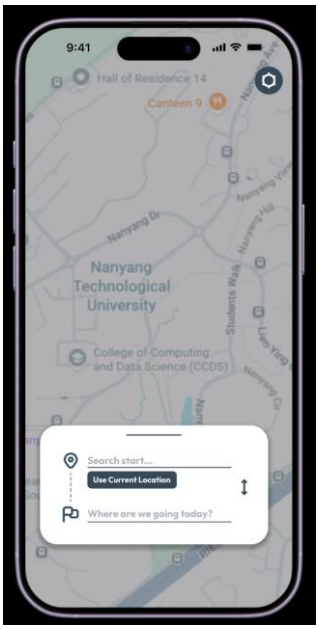
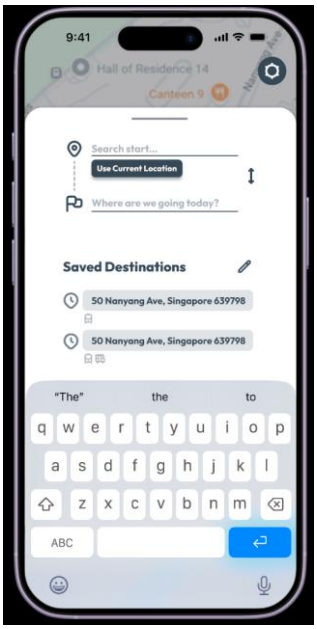
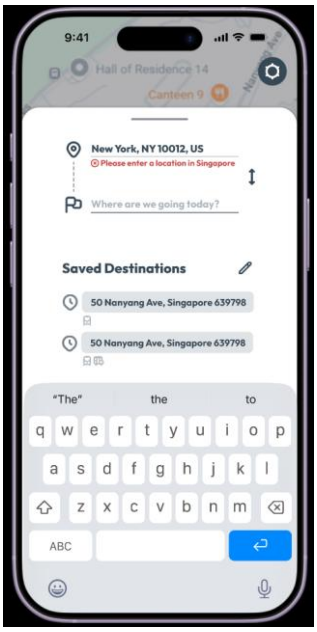
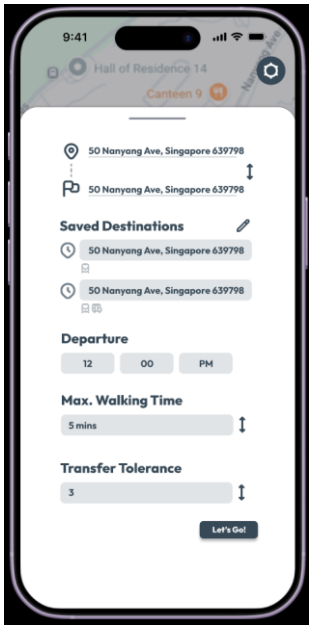
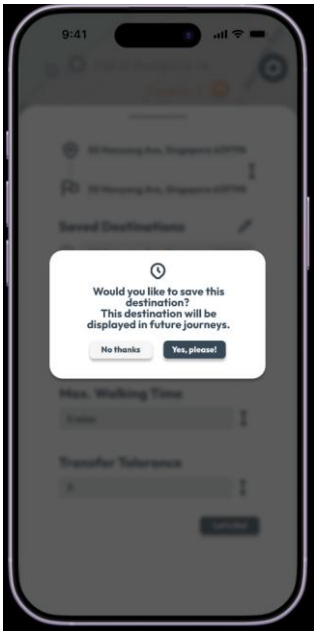
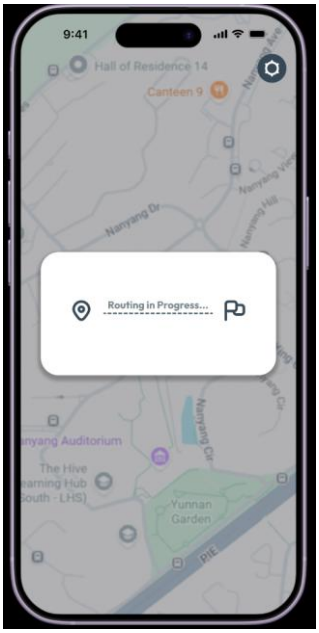
Use Case ID:	23		
Use Case Name:	Change Settings		
Created By:	Ong Zhi Jie	Last Updated By:	Ong Zhi Jie
Date Created:	11/2/2026	Date Last Updated:	11/2/2026

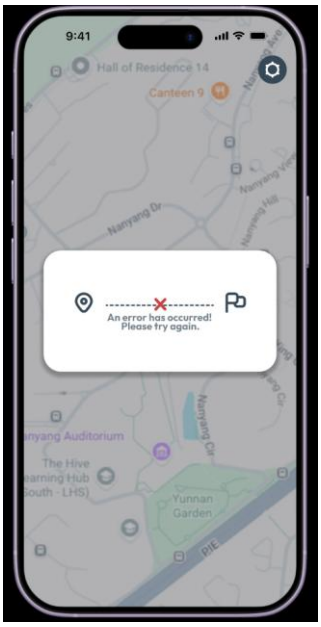
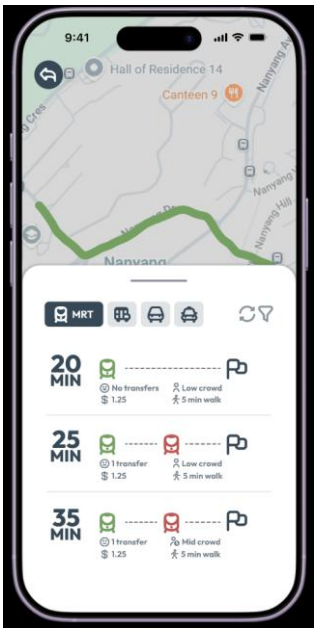
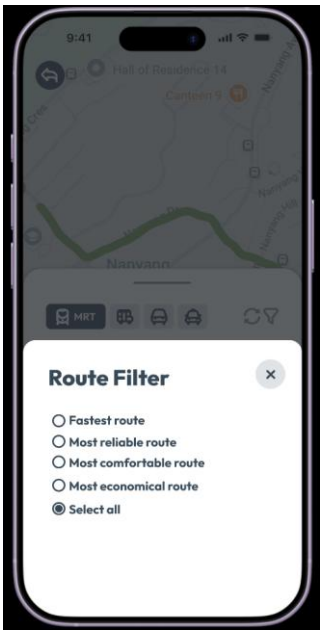
Actor:	User
Description:	Allows the user to modify global application configurations and system-level preferences
Preconditions:	The user has accessed the settings menu
Postconditions:	Settings are changed or remain the same (discarded changes)
Priority:	Medium
Frequency of Use:	Low
Flow of Events:	1. User navigates to the "Settings" interface. 2. System displays current configuration options (e.g. units of measure). 3. User modifies the desired settings 4. User initiates the save process (**Extend: Save Settings**)
Alternative Flows:	**Cancel Changes** User exits the settings menu without confirming; system discards all modifications.
Exceptions:	NIL
Includes:	None
Special Requirements:	NIL
Assumptions:	NIL
Notes and Issues:	NIL

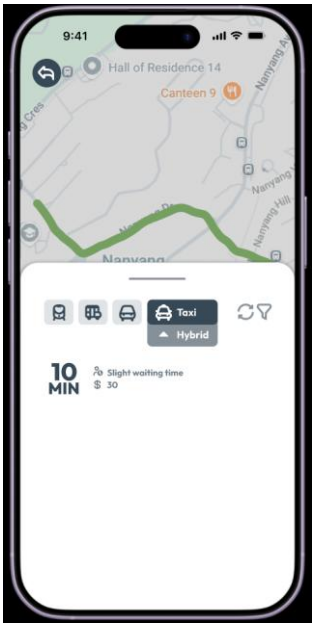
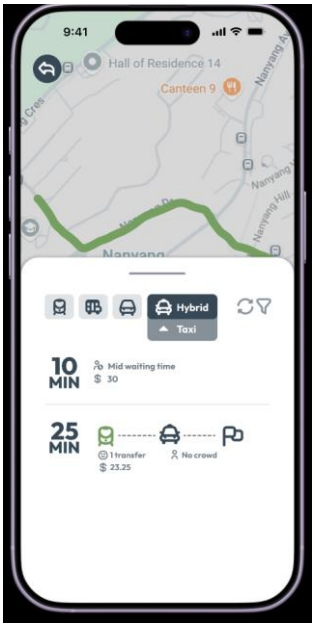
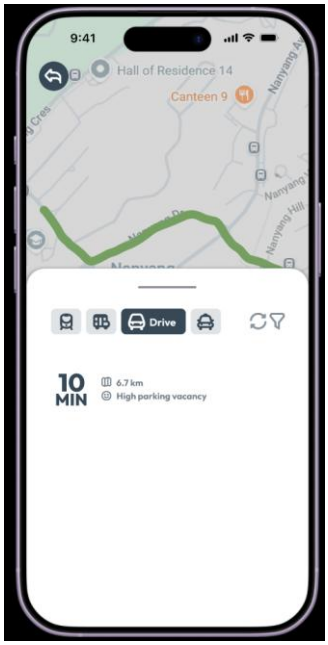
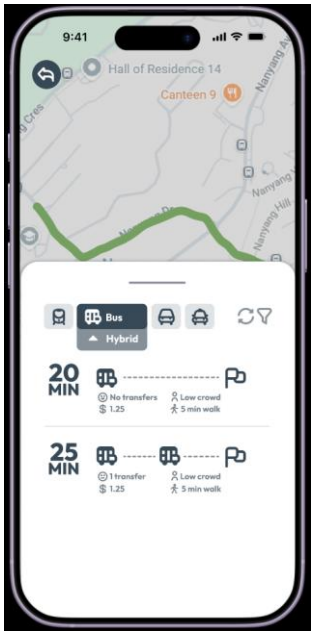
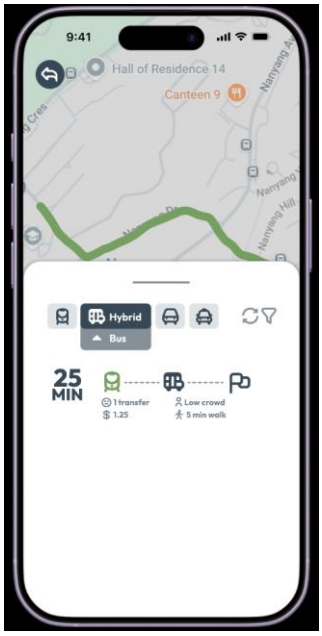
Use Case ID:	24		
Use Case Name:	Save Settings		
Created By:	Ong Zhi Jie	Last Updated By:	Ong Zhi Jie
Date Created:	11/2/2026	Date Last Updated:	11/2/2026

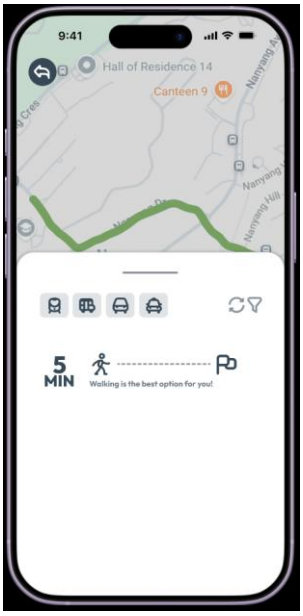
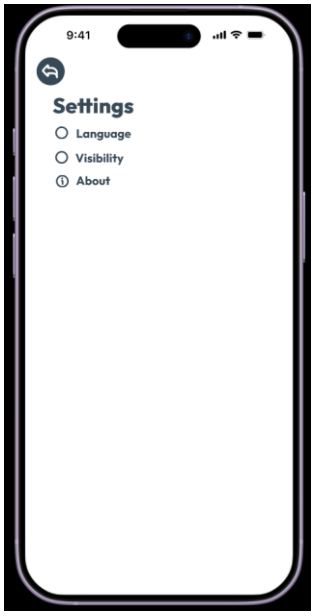
Actor:	User
Description:	Commits modified application configurations to the system's persistent storage
Preconditions:	The user has modified one or more global settings
Postconditions:	Configuration changes are stored and active for the user's account
Priority:	Medium
Frequency of Use:	Low
Flow of Events:	1. System receives updated configuration parameters. 2. System overwrites existing settings in the database. 3. System notifies the user of the successful update.
Alternative Flows:	NIL
Exceptions:	**Write Error** Database connection fails; the system notifies the user and retains the pending changes in the interface for a retry.
Includes:	None
Special Requirements:	NIL
Assumptions:	NIL
Notes and Issues:	NIL

Initial UI Mockup Diagram

		
The application requests users to input their current location & destination.	The application displays user's saved destinations and modes of transport.	If an address that is not within Singapore is inputted, application displays an error message.
		
Application allows users to enter their preferred departure timing (starts from current timing), maximum walking time and transfer tolerance of their travel.	After 'Let's Go' is pressed, application asks user if they would like to save destination.	Loading screen when application is calculating all possible routes.

		
In the event of an issue, an error message will pop up and prompt user to try again.	After calculating all possible routes, routes for each transport mode will be made available, along with: • Travel time • Line transfers • Total price • Crowdedness of stations • Walk time Users can click the rerun icon to retrieve the latest updated routes.	Users will be able to personalize route filters based on: • Fastest route • Most reliable route (least transfers) • Most comfortable route (least crowd) • Most economical route (least amount of money spent) • All of the above

		
When users click the 'taxi' option, there will be an option to choose hybrid or taxi only along with: • Travel time • Taxi waiting time • Taxi fare	If the 'hybrid' option is clicked, the application provides routes where transfers of transportation mode is possible.	Users will be able to personalize route filters based on: • Fastest route • Most reliable route (least transfers) • Most comfortable route (least crowd) • All of the above • Their budget
		

If the 'drive' option is clicked, the total time taken, the screen will display: • Travel time • Distance to drive • Parking lot availability of destination	When the user clicks the 'bus' option, there will be an option to choose hybrid or bus only along with: • Travel time • Bus transfers • Total price • Crowdedness of buses • Walk time	If the 'hybrid' option is clicked, the application provides routes where transfers of transportation mode is possible.
		
If walking is the only available way to get to a destination, the application would let users know.	The settings page allows users to customize language and visibility.	